



Internship Final Presentation

Maturing Mars Small Satellite Missions to Inform Bounds for Delivery Tug

September 3, 2024

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Reviewed and determined not to contain CUI.
JPL URS clearance number: CL#25-0243



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California Institute of Technology

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Table of Contents

- About me
- Introduction
- Task Overview & Objectives
- Approach
- Key Assumptions
- My Contributions
- Conclusions
- Lessons Learned
- Future Goals
- Acknowledgements

About Me

School: University of California San Diego (UCSD) Class of 2025

Major: Aerospace Engineering with a focus in Space Applications

Future Plans: Remaining at UCSD for a Masters degree in Mechanical Engineering in their BS/MS program -- Currently exploring specialization options



Outside of JPL and courses...

- Students for the Exploration and Development of Space (SEDS)
 - Lead Recovery Engineer of 20ft liquid bi-prop rocket
 - Director of Internal Affairs
 - Female Engineering Mentorship (FEM) Program Co-lead
- **Interests:** Reading, exploring, crochet, embroidery, working on cars

Introduction

- International Mars Exploration Working Group (IMEWG)
 - 30 Nations working to collaborate on strategic Mars initiatives of mutual interest to share knowledge, costs, risks, opportunities and benefits for humankind.
- Lower-Cost Mission Working Group
 - Group under IMEWG performing Earth2Mars Study
- Earth2Mars Study
 - International survey of lower-cost Mars mission concepts
 - Hypothetical international multi-mission concept combined on one carrier vehicle
 - Goals to produce a database of Mars mission concepts, concept presentations, and a possible paper.

Task Overview & Objectives

Task: Create and mature 2 low-cost, SmallSat Mars mission concepts. With that information, generate bounding values for a Space Tug that will transport these and similar missions from Earth to Mars.

Task Objectives:

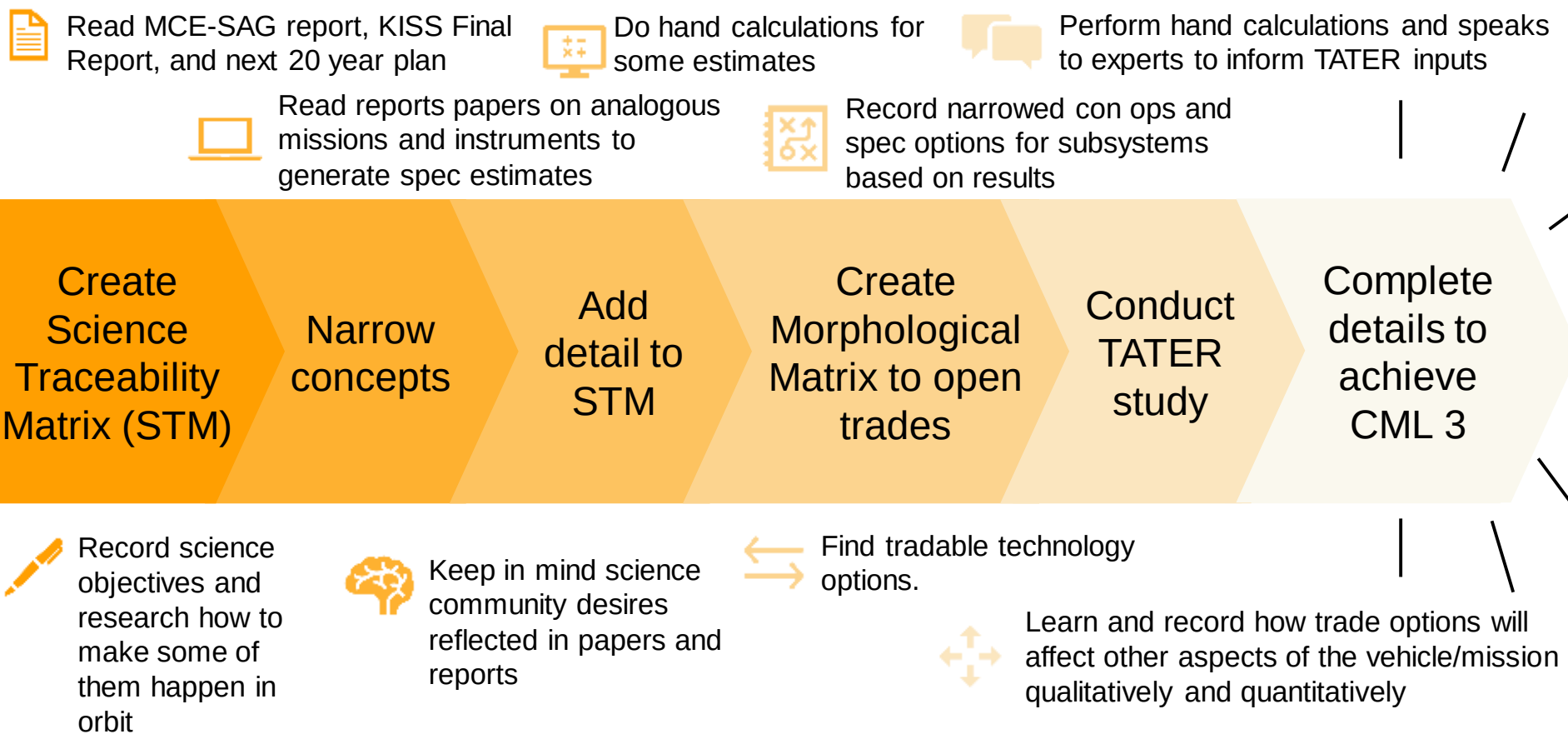
- Conduct a study to mature 2 Mars satellite mission concepts to be manifested on a delivery space tug with other missions
 - Use MCE-SAG report to record Decadal science objectives for potential missions
 - Use MCE-SAG report, KISS study, and analogous missions to ensure high feasibility
- Explore trade spaces for each concept
 - Assess sensitivities in critical subsystems
- Generate bounds for delivery tug
 - Use concepts from international partners with my own to visualize strawman mission
 - Determine adjustability of drop off orbit
 - Explore drop-off route options



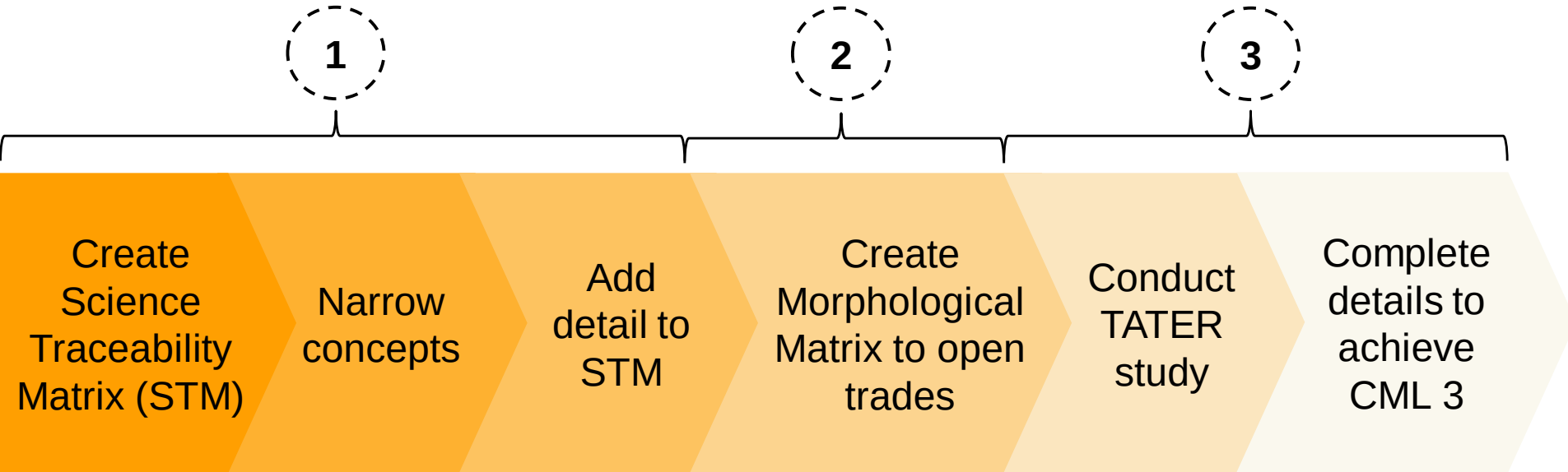
Approach for Creation of SmallSat Mission Concept



Approach for Creation of SmallSat Mission Concept



Approach for Creation of SmallSat Mission Concept



1

Create Science Traceability Matrix (STM)

Narrow concepts

Add detail to STM

Key Assumptions and Constraints

- Small Satellites will be dropped off by a Space Tug
 - Estimated total carried mass will be 1250 kg across all missions on manifest
 - Tug will provide relay services to Earth
 - 300 km x 300 km polar drop off orbit
- Connection with tug must be standard (ESPA ring, CubeSat deployer)
- Concept missions must be ESPA-Class SmallSats (400 kg or less), CubeSats (up to 12U), or hosted payloads
 - Minimum mission duration is 1 Mars Year
 - Objective of performing decadal-level science

My Contributions – Science Traceability Matrix

	Description	Scientific Goals Met	Scientific Objectives	Physical Parameters	Observables	Analogy Instruments	Architectures and Instruments (TRL#, SWaP, avg data rate)	Technical Notes	Notes
High Feasibility, High Priority: Most Viable + Higher Precedence Hard Lander -- (Not in any particular order) -- YELLOW HIGHLIGHT INDICATES DESIRED DEPTH OF STUDY									
Orbiter		Planetary Evolution goal of MCE-SAG OWL 4.3, 5.1, 5.2, 5.3, 5.6 MEPAG Goal III A, B	Determine the composition and petrologic evolution of the early martian crust [PE1]	// Resolution better than current resources [MGS][alt="450km"], [ODV][alt="400km"], [MRO][alt="250-316km"], [CRISM] [alt="100m"]	// between satellite range values // range values between satellites	GRAIL, GRACE, GRACE-FO,	[P] Microwave ranging (9, #, #, #, #, 2kbits/s dnlink) [P] Laser ranging interferometer (8, #, #, #, #) GRAIL dry mass w/ antenna, prop, and attitude control = 136.2kg atom interferometer gradiometer (47, #, #, #, #) in situ gravimeter (9, #, 0.75kg, #, #) [P] LIDAR (#, #, 24x.228x.25m, 3.7kg, 17W, 1Hz), (#, #, 2.37kg, 6.8W, 1630kpps)	GRAIL used Lockheed Martin Experimental Small Sat (XSS-111) bus // near circular sun synch orbit (low altitude) // pointing matters for imagers depending on resolution desired	From "Mars Small Spacecraft Studies: Overview" (mentions that these can be achieved through aerostationary orbit) [mentioned in MCE-SAG pg 61] // From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to have "higher precedence") // Ask someone about the current state of the art
	SWIR/TIR Spectrometry	Planetary Evolution goal of MCE-SAG OWL 5.1, 5.2 MEPAG Goal III B1, B2	Determine the composition and petrologic evolution of the early martian crust [PE2]			THEMIS, CRISM (TRL9), HiRISE, MarCO	[P] Orbital VNIR/SWIR multispectral spectrometer (#, #, 19kg, #, #) NIR () [P] Orbital VNIR/SWIR hyperspectral spectrometer (CRISM) (9, #, #, 32.92kg, #, #) [P] (for resolution) Orbital Sounding Radar (high, #, #, #, #) [P] Orbital VNIR/SWIR hyperspectral spectrometer (CRISM) (9, #, #, 32.92kg, #, #) [P] in situ mineralogy (VNIR+TIR particle size, #, #, #, #) [P] (for resolution) Groundpenetrating radar (GPR) (2, #, #, #, #, 2.5W, #, #)	Orbital (can have 2nd more in depth mission with rotorcraft) Potential combination with EE1 // Potential equatorial orbit around dichotomy?	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to be "most viable") (determined by report to have "higher precedence")
	Orbital Sounding Radar over key ice regions (high resolution)	Recent Climate Evolution goal of MCE-SAG OWL 5.4, 6.2 MEPAG Goal II A2, B1, B2, B3, III A1 NEX-SAG Findings 4, 5 ICE-SAG Findings 2, 5 I-MIM Finding 5	Determine how Mars's climate has changed over orbital time scales as recorded in global ice stratigraphies [RC1]			CRISM, Min-TES, ICESat, ICESat-2, CryoSat-2	[P] (for resolution) Orbital Sounding Radar (high, #, #, #, #) [P] Orbital VNIR/SWIR hyperspectral spectrometer (CRISM) (9, #, #, 32.92kg, #, #) [P] in situ mineralogy (VNIR+TIR particle size, #, #, #, #) [P] (for resolution) Groundpenetrating radar (GPR) (2, #, #, #, #, 2.5W, #, #)	Potential near circular SS polar orbit // pointing matters for imagers depending on resolution desired	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to be "most viable") (determined by report to have "higher precedence" but with landers)
	climate orbiter with infrared spectrometer, weather imaging system, wind/speed profile instrument	Dynamic Modern Environments goal of MCE-SAG OWL 6.3 MEPAG Goal II A1, A2, IV A1, B3 Terrae Novae Weather Network	Understand atmosphere/climate variability including the timing of dust storms [DM2]			Lunar Flashlight (current), MarCO, GRIFFEX	Meteorology Station (p, T, surface wind, also humidity) (#, #, #, #, #) Temperature (9, #, #, 0.02kg, #, #) Winds (5, #, 0.5kg, #, #) Weather camera (9, #, #, 0.835kg, #, #) [P] (for resolution) Mars Doppler Wind and Temperature Sounder (MDWTS) (4, #, #, #, #) Microwave imager/sounder SABER(?) MCS	orbiter (Can have 2nd more in depth mission with hard lander with 4D dust and weather capabilities) // Potential SS polar orbit (Mars Climate orbiter intended orbit, OCO-2) // pointing matters for imagers depending on resolution desired // potential for aerostationary constellation	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to be "most viable") (determined by report to have "higher precedence" but Mars has had weather monitored by polar or quasi-polar orbiters)
	UV Sounder to probe upper atmospheric chemistry and escape	Dynamic Modern Environments goal of MCE-SAG OWL 6.5 MEPAG Goal II A4, IV A1	Determine how lower atmospheric dynamics escape to space [DM4]			Lunar Flashlight (current), MarCO, GRIFFEX	[P] Orbital UV imager (9, #, #, 9kg/27kg/18kg, #, #) Microwave imager/sounder	Elliptical orbit (Ulysses style orbit) (MAVEN) (let scientist determine based on need) // pointing matters for imagers depending on resolution desired	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to be "most viable") (determined by report to have "higher precedence")
Hard Lander	Landed Siesmic Network	Planetary Evolution goal of MCE-SAG OWL 4.3, 5.1, 5.2, 5.3, 5.6 MEPAG Goal III A, B	Determine the global structure of the martian crust and lithosphere, including the origin and nature of the hemispheric dichotomy [PE1]			InSight	SEIS Seismic network (#, #, 10kg, #, #)	lander(s)	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations mentioned on page 25 of MCE-SAG report as an option for fulfilling the PE1 desired investigation
	Temperature, 2D wind, radiative measurements, pressure	Dynamic Modern Environments goal of MCE-SAG MEPAG Goal II A1, A2, A4, IV A1, B3 Terrae Novae Weather Network NEX-SAG Finding 9, 12 ICE-SAG Finding 3	Understand and characterize the controlling factors for surface weather on Mars [DM8]	// Instruments must be able to survive hard lander impact	// temperature // wind speed (maybe direction) // pressure	Pathfinder, MET on MPL used for mass and power for "Weather Station"	Temperature (9, #, #, 0.02kg, #, #) Winds (5, #, 0.5kg, #, #) [P] (for resolution) Mars Doppler Wind and Temperature Sounder (MDWTS) (4, #, #, #, #) Weather Station (from SHIELD Team X study) (#, #, 3.9kg (MEV), 24.3W (MEV), #, #)	Lander (helps facilitate DM1) // If using SHIELD, add 100kg and payload for shield is 6kg?	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to have "higher precedence") mentions desire for multiple small-med sized landers, mentions that optimal location placement would require an "observing system simulation experiment"

Reviewed and determined not to contain CUI.

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My Contributions – Science Traceability Matrix

	Description	Scientific Goals Met	Scientific Objectives	Physical Parameters	Observables	Analogy Instruments	Architectures and Instruments (TRL#, SWaP, avg data rate)	Technical Notes	Notes
High Feasibility, High Priority: Most Viable + Higher Precedence Hard Lander -- (Not in any particular order) -- YELLOW HIGHLIGHT INDICATES DESIRED DEPTH OF STUDY									
Orbiter	Considered Concepts								
		Planetary Evolution goal of MCE-SAG	Determine the composition and				[P] Orbital VNIR/SWIR multispectral spectrometer (#, #, 19kg, #, #) NIR ()	Orbital (can have 2nd more in depth mission with rotorcraft) Potential	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by
	SWIR								
	Orbital key resolu	Planetary Evolution goal of MCE-SAG OWL 4.3, 5.1, 5.2, 5.3, 5.6 MEPAG Goal III A, B	Determine the global structure of the martian crust and lithosphere, including the origin and nature of the hemispheric dichotomy [PE1]	// Resolution better than current resources (MGS[alt=~450km], OD[alt=~450km], MMR[alt=~450km], 100km anomalies & 12.5km resolution crust thickness // GRACE alt was 483-508km (LEO is below 2000km)	// between satellite range values // range rate values between satellites	GRAIL, GRACE, GRACE-FO	[P] Microwave ranging (9, #, #, #, 2kbits/s dnlink) [P] laser ranging interferometer (8, #, #, #) [P] LIDAR dry mass w/ antennae, prop, and attitude control = 136.2kg atom interferometer gradiometer(47, #, #, #) in situ gravimetry (9, #, 0.75kg, #, #) [P] LIDAR (#, .24x.228x.25m, 3.7kg, 17W, 1Hz), (#, #, 2.37kg, 6.8W, 1630kbps)	GRAIL used Lockheed Martin Experimental Small Sat (XSS-11) bus // near circular sun synchronous polar orbit (low altitude) // pointing matters for imagers depending on resolution desired	From "Mars Small Spacecraft Studies: Overview" (mentions that these can be achieved through aerostationary orbit) [mentioned in MCE-SAG pg 61] // From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to be "most viable") (determined by report to have "higher precedence") // Ask someone about GRACE and GRACE-FO and what the current state of tech is
	clima infra weather imaging system, wind/speed profile instrument	Dynamic Modern Environments goal of MCE-SAG MEPAG Goal II A1, A2, A4, IV A1, B3 Terra Nova Weather Network NEX-SAG Finding 9, 12 ICE-SAG Finding 3	Understand and characterize the controlling factors for surface weather on Mars [DM8]				Temperature (9, #, 0.02kg, #, #) Winds (5, #, 0.5kg, #, #) Mars Doppler Wind and Temperature Sounder (MDWTS) (4, #, #, #, #) Weather Station (from SHIELD Team X study) (#, #, 3.9kg (MEV), 24.3W (MEV), #)	Lander (helps facilitate DM1) // If using SHIELD, add 100kg and payload for shield is 6kg?	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to have "higher precedence") mentions desire for multiple small-med sized landers, mentions that optimal location placement would require an "observing system simulation experiment"
Hard Lander	UV Sounder to probe upper atmospheric chemistry and escape	Dynamic Modern Environments goal of MCE-SAG OWL 6.5 MEPAG Goal II A4, IV A1	Determine how lower atmospheric dynamics escape to space [DM4]			Lunar Flashlight (current), MarCO, GRIFFEX	[P] Orbital UV imager (9, #, 9kg/27kg/18kg, #, #) Microwave imager/sounder	Elliptical orbit (UARS), circular orbit (MAVEN) [let scientist determine based on need] // pointing matters for imagers depending on resolution desired	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations (determined by report to be "most viable") (determined by report to have "higher precedence")
	Landed Sismic Network	Planetary Evolution goal of MCE-SAG OWL 4.3, 5.1, 5.2, 5.3, 5.6 MEPAG Goal III A, B	Determine the global structure of the martian crust and lithosphere, including the origin and nature of the hemispheric dichotomy [PE1]			InSight	SEIS Seismic network (#, #, 10kg, #, #)	lander(s)	From MCE-SAG Final Report Appendix 5 Science Objectives and Investigations mentioned on page 25 of MCE-SAG report as an option for fulfilling the PE1 desired investigation

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2

Create Morphological Matrix to open trades

My Contributions – Considered Potential Architectures

	Gravity Measuring Orbiter		Dust and Weather Hard Lander			
Instrument	2-way microwave ranging (modified LGRS)	2-way laser ranging interferometer (demonstrated on GRACE-FO)	MET (Meteorology Package)	ATMIS (Atmospheric & Meteorological Instrumentation System)	PAMI Instrument Suite	ASI/MET (Atmospheric Structure Instrument/ Meteorology Package)
Instrument Specs	TRL 9 // 14.9 kg // 40.2 W	High TRL //	TRL 8 // 3.2 kg // 5W // MVACS, Mars Polar Lander	0.855kg // 0.43W //NETLANDER	less than or equal to 6kg // less than or equal to 120W // in SHIELD concept paper	High TRL // 1.3kg // 9.3 W // Pathfinder
Spacecraft (s/c) Configuration	2 identical s/c		1 Lander		Multi-lander network	
Orbit	Polar SSO		N/A, must land within 30° South and 30° North latitude			
Altitude	200km	250km	N/A, must land within 30° South and 30° North latitude			
Power System	Solar Panels & Batteries		Batteries only	RTG or equivalent	Solar Panels + batteries	
Operational Lifetime	1 Martian Year	>1 Martian Year	1 Martian Year		>1 Martian year	

Reviewed and determined not to contain CUI.

My Contributions – Considered Potential Architectures

	Dust and Weather Hard Lander	
Instrument	MicroMED	MEDA
Instrument Specs	0.5kg // 3W // 2U	TRL 9 // 5.5 kg // 17 W // Perseverance
Spacecraft (s/c) Configuration	1 Lander	Multi-lander network
Orbit	N/A, must land within 30° South and 30° North latitude	
Altitude	N/A, must land within 30° South and 30° North latitude	
Power System	Solar Panels + Batteries	
Operational Lifetime	1 Martian Year	>1 Martian Year

- Continuation of Dust and Weather Hard Lander instrument options
 - Instruments are a key trade for this mission concept
 - Instruments must survive impact and be operational even after bounce or in multiple orientations

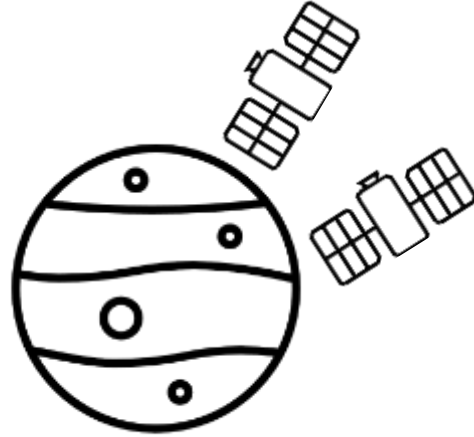


Conduct TATER study
Complete details to achieve CML 3

My Contributions

Mission Concepts

My Contributions – SmallSat Mission Concept 1

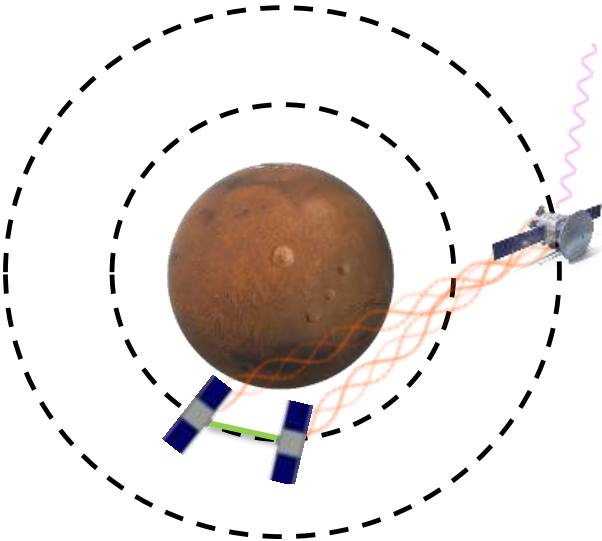


Gravity Measurement Orbiter

Mission Overview

Nominal Design – Modeled in TATER

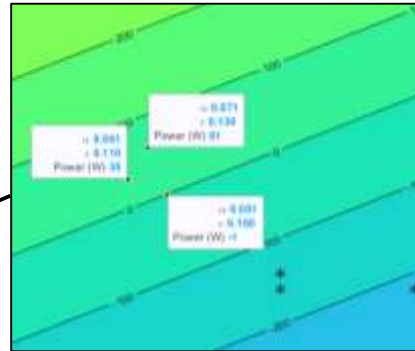
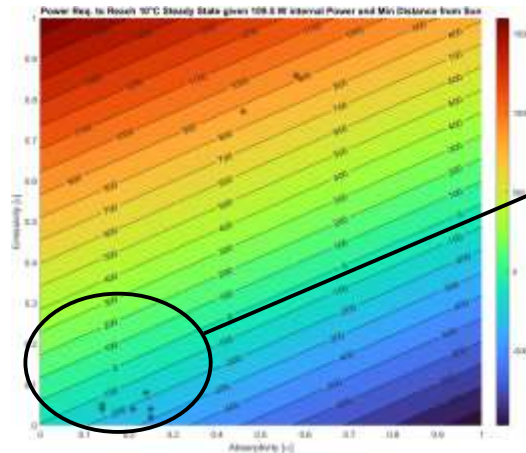
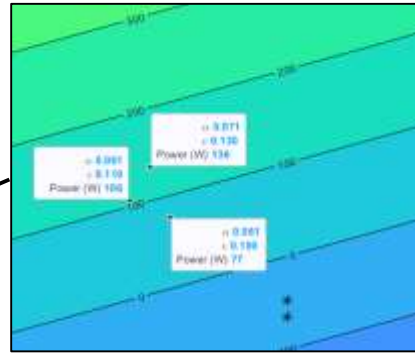
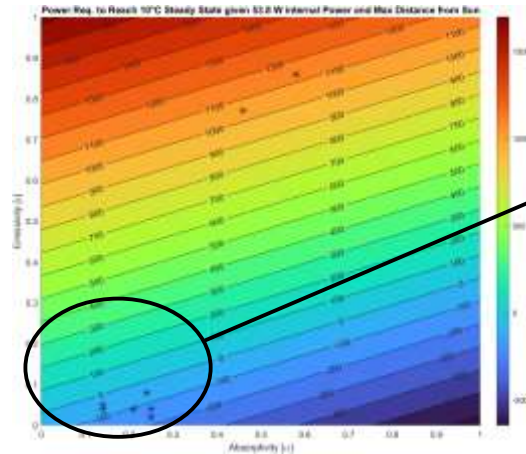
Mass	Total Dry Predicted	87 kg
	Total Wet Allowable	106 kg
Power Modes	Checkout	215 W
	Science	181 W
	Maneuver	160 W
	Telecom	209 W
	Battery Post-Deployment	164 W
Power	Thermal	106 W
	Instrument	15 W
Propulsion	Monopropellant system	65 m/s of dv
CDS	Sphinx	
Telecom	IRIS	
ACS	Analogous to INCUS	



Areas of interest:

- Difference in mass between modeled and analogous missions
- Thermal power effects
- Delta v effects

Trades Explored – Thermal

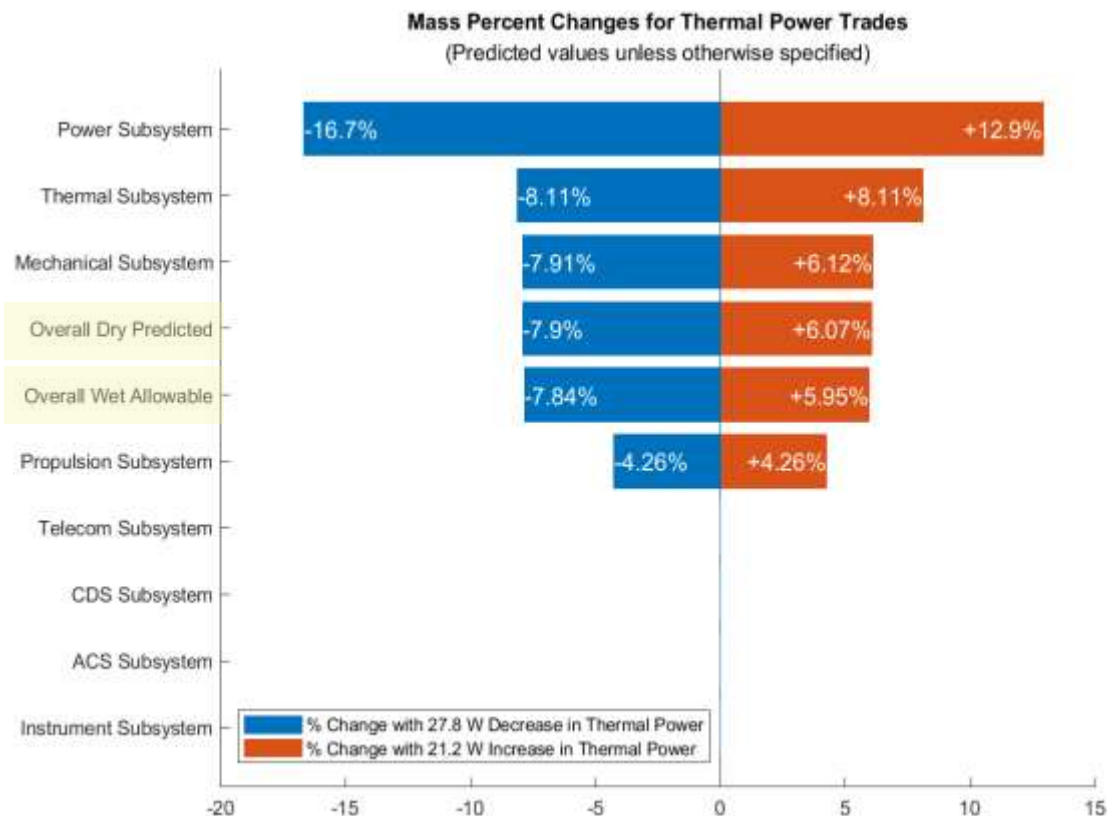


Single node thermal analysis for steady state temperature of satellite bus

- 2 power and orientation modes that provide bounding characteristics to heater power
- Allowed pinpointing of Absorptivity and Emissivity values that correspond to desired heater power needs
 - Allows targeting for a thermal coating, insulation, or paint
- Visualized the goal of cold biasing the spacecraft without massive power needs

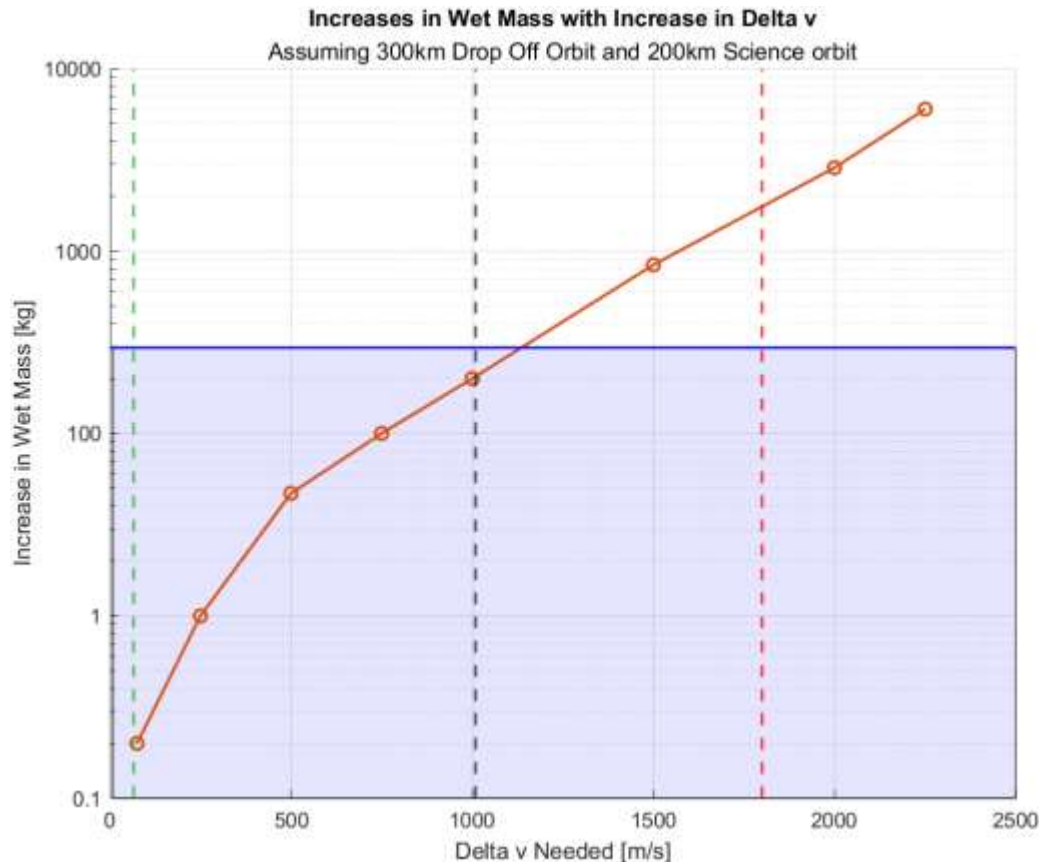
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Trades Explored - Thermal



- Note: TATER is only verified for Element level mass estimates (Overall Dry Predicted and Overall Wet Allowable)
 - Still useful to see the most affected subsystems
 - Decreasing necessary thermal power by 28 W produces an almost 8% decrease in mass
 - Reveals sensitivities in coatings
 - Allows a range of usable heat capabilities all resulting in a wet allowable mass <115 kg

Trades Explored – How Orbit Δv Requirements Affect Mass



**For Chemical Propulsion system using monopropellant

- Science inclination has limited range
- Altitude drop is not as sensitive
 - Not in graph: 1000 km altitude drop with no plane change requires 400 m/s Δv
- Delta v needs are limited to ~1150 m/s according to this plot
 - Including considerations for correctional maneuvers and station keeping
 - ***Provides science orbit limitations and/or drives necessities for tug drop off orbit if more missions have similar needs.***

Mass Rollup of Nominal Design and Comparison with GRAIL

	GRAIL Flight Values	Mars Gravity Orbiter TATER Predicted Mass Values	% Change
Total Dry	201 kg	87.3 kg	-57%
Total Wet*	307 kg	90.3 kg	-71%
Instrument	17.7 kg	14.9 kg	-16%
Attitude Control Systems	14.7 kg	2.1 kg	-86%
Command and Data Subsystem	11.6 kg	1.3 kg	-89%
Mechanical	89.9 kg	27.8 kg	-69%
Power	27.2 kg	24.0 kg	-12%
Propulsion (dry)	28.1 kg	4.7 kg	-83%
Thermal	8.9 kg	3.7 kg	-58%
Telecom	3.2 kg	4.2 kg	31%

***This type of mass drop has also been seen in other concept submissions*

** TATER prop sizing against total wet allowable (total wet allowable -65.54% difference), (total dry basic @ 67.2kg is -78.11% difference)*

Reviewed and determined not to contain CUI.

Mass Rollup of Nominal Design and Comparison with GRAIL

How is the mass so different?

- *Attitude Control Systems*
 - Mars Gravity Orbiter components can be SmallSat rated because of drop off
 - Reaction wheels may grow to extend the momentum capacity
 - Due to planned science operations and frequency of desaturation
- *Command and Data Subsystem*
 - Mars Gravity Orbiter is using Sphinx (also used on Lunar Flashlight)
 - Much smaller than other processors
 - Sized for CubeSats
- *Propulsion*
 - Drop off Delivery tug means propulsion needs are reduced greatly
 - Drop off orbit is currently planned to be very close to desired science orbit
- *Mechanical*
 - Reduced by everything else being reduced

My Contributions – SmallSat Mission Concept 2



Dust and Weather Hard
Lander

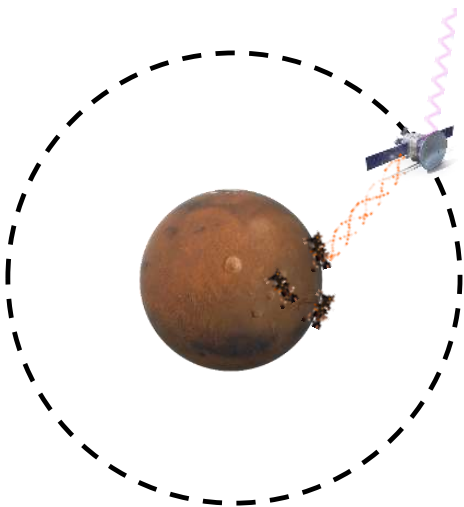
Background on SHIELD Vehicle

- Hard Lander vehicle concept that uses an accordion-like structure to hit the Martian surface at 70 m/s
- Houses a Warm Electronics Box (WEB) capable of supporting instruments up to 120 W between -20 and 0 °C
 - Powered by solar panels on lander
- Assumed to house instruments up to 6 kg in a 30 cm x 30 cm x 10 cm usable space where total includes anything not included in the WEB, deployables, booms, actuators, and the instrument
 - The mass and volume values are adjustable but may require additional power and heating to accommodate

Mission Overview

Nominal Values for Max Capacity of the Vehicle – Modeled in TATER

Mass	Total Dry Predicted	58.1 kg + 255 kg
	Total Wet Allowable	78.3 kg + 255 kg
Power Modes	Checkout	322 W
	Science	291 W
	Maneuver	178.2 W
	Telecom	318.5 W
	Battery Post-Deployment	186.5 W
Power	Thermal	127 W
	Instrument	120 W
Propulsion	Monopropellant system	70 m/s of dv
CDS	Sphinx	
Telecom	IRIS	
ACS	Analogous to INCUS	



Areas of interest:

- Different instruments with less mass and power needs
- Instrument configurations that are independent of orientation

Trades Explored - Instrumentation

Instrument	MET (Meteorology Package)	ATMIS (Atmospheric and Meteorological Instrumentation System)	PAMI Instrument Suite	MEDA (Mars Environnemental Dynamics Analyzer)	ASI/MET (Atmospheric Structure Instrument/Meteor ology Package)*
TRL	8	Medium	Medium	9	9
Mass	3.2	0.855	6	5.5 kg	1.3
Power	5	0.43	120	17 W	9.3
Heritage	MVACS, Mars Polar Lander	NETLANDER	Studied in SHIELD concept overview paper	Perseverance	Pathfinder
Notes			Sonic Anemometer, Laser Anemometer, Laser Spectrometer, Pressure Sensor, Flux Radiometers, Temperature Sensors, Cameras, Saltation Sensor	11 Mb of return data // 5 Air temperature sensors, Radiation and dust sensor, Relative humidity sensor, Thermal infrared sensor, 2 Wind sensors, Instrument control unit and pressure sensor	

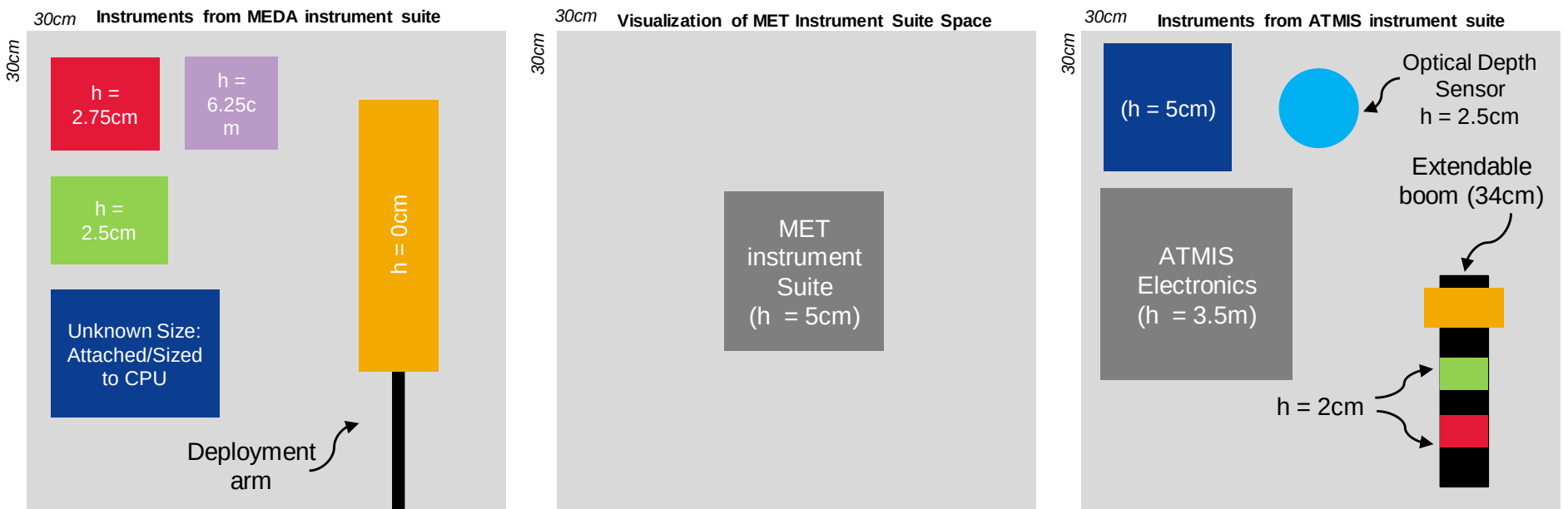
Trades Explored – Stowed Configurations in 30x30x10cm

Must have instruments: Nice-to-have Instruments:

- Temperature sensor
- Pressure sensor
- Wind sensor
- Thermal IR Sensor
- Relative humidity Sensor

Notes for Future Study:

- Which Instruments require extension past the vehicle
- Padding techniques for impact



*** For all Configurations, size of instrument is based on instruments found in trade study either separated from suites or entire suites*

Reviewed and determined not to contain CUI.

My Contributions

Bringing Everything Together

Presentation to Low-Cost Mission Working Group International Partners

Gravity Measurement Orbiter

- Science Objective:** Determine the global structure of the Martian crust and lithosphere, including the origin and nature of the hemispheric dichotomy.
 - (MCE-02) Map Planetary Evolution Unit 1: High Fidelity + High Priority + Must Have reason concept (detailed reason concept) in RDR study
- Observables:** Gravity field map. Spherical harmonic degree of 190-200 or better (53km spatial resolution).
- Measurements:** Ranges and range rate values between two orbiters caused by mass differences on Mars.

Mission Requirements

[illegible]

Resources/Accommodations From Tag

Host Platform	32-bit, x86	
Operating System	Windows 95/98/NT	Windows 95/98/NT
Processor	Intel Pentium 486/586/686	Intel Pentium 486/586/686
Memory	16 MB (minimum)	16 MB (minimum)
Storage	10 MB (minimum)	10 MB (minimum)
Network	10/100 Mbps Ethernet	10/100 Mbps Ethernet
Security	Windows NT/2000/XP/2003/2008/2012/2016/2019/2022	Windows NT/2000/XP/2003/2008/2012/2016/2019/2022
Other	Windows 10/11/12/13/14/15/16/17/18/19/20/21/22/23/24/25/26/27/28/29/30/31/32/33/34/35/36/37/38/39/40/41/42/43/44/45/46/47/48/49/50/51/52/53/54/55/56/57/58/59/60/61/62/63/64/65/66/67/68/69/70/71/72/73/74/75/76/77/78/79/80/81/82/83/84/85/86/87/88/89/90/91/92/93/94/95/96/97/98/99/100	Windows 10/11/12/13/14/15/16/17/18/19/20/21/22/23/24/25/26/27/28/29/30/31/32/33/34/35/36/37/38/39/40/41/42/43/44/45/46/47/48/49/50/51/52/53/54/55/56/57/58/59/60/61/62/63/64/65/66/67/68/69/70/71/72/73/74/75/76/77/78/79/80/81/82/83/84/85/86/87/88/89/90/91/92/93/94/95/96/97/98/99/100

Technical Readiness

Current TR	High, new architecture
Next-Gen Device	GRAS, GRACE, GRACE-PS Program will be for orbit transfer from drop off and start-up/stop (1 GRAS, reference for analogous instrument, initial testing)

Dust and Weather Hard Lander

- **Science Objective:** Understand and characterize the controlling factors for surface weather on Mars.
 - JSCC-SAG report Synoptic Mars Goals & Higher Prioritization
- **Observables:** Surface-temperature measurements and variation of surface weather, dust cycles
- **Measurements:** Near surface horizontal and vertical wind speed, surface pressure, and temperature.
 - Network Mars weather sensors: other landers, low orbits
 - Additional science potential with RF signals through atmosphere
 - Easy to adapt low power "batter" instruments to maximize power

Mission Requirements

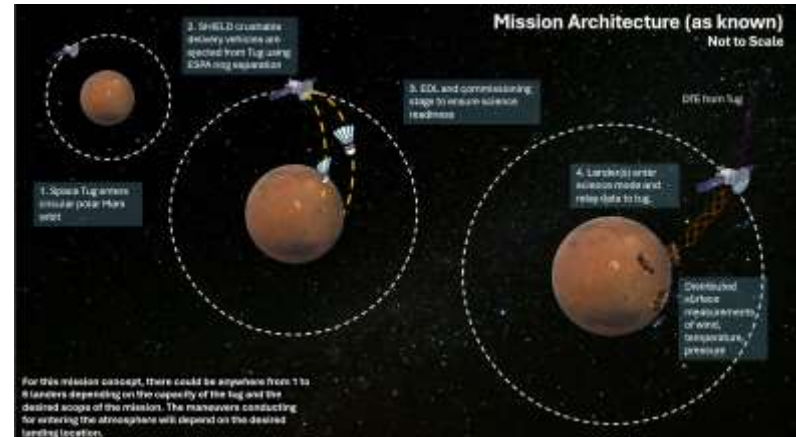
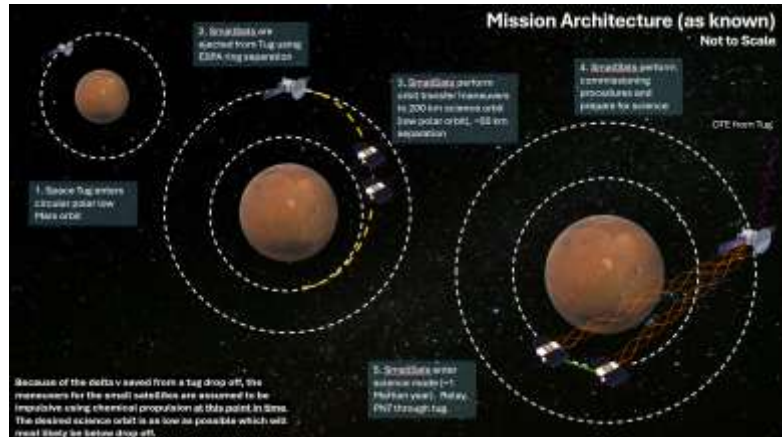
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Resources/Accommodations From Tug

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Technical Readiness

Current TRL	N/A
Technology Maturity	Instrumental: Prototypes, initial technology development for high volume.
	Based on use of the Fuel Injector Delivery available products (FIDEL), proved capability to measure horizontal and vertical wind speed.



Space Tug Trade Space Analysis

Areas of Interest for Space-tug: Propulsion type, Relay Capabilities, Final Relay Orbit

Propulsion Type	
Chemical	Solar Electric
Pros	
Shorter cruise duration	High Efficiency
Higher Thrust	Low Mass
Cons	
High Mass	Longer cruise duration
Low efficiency	Lower Thrust

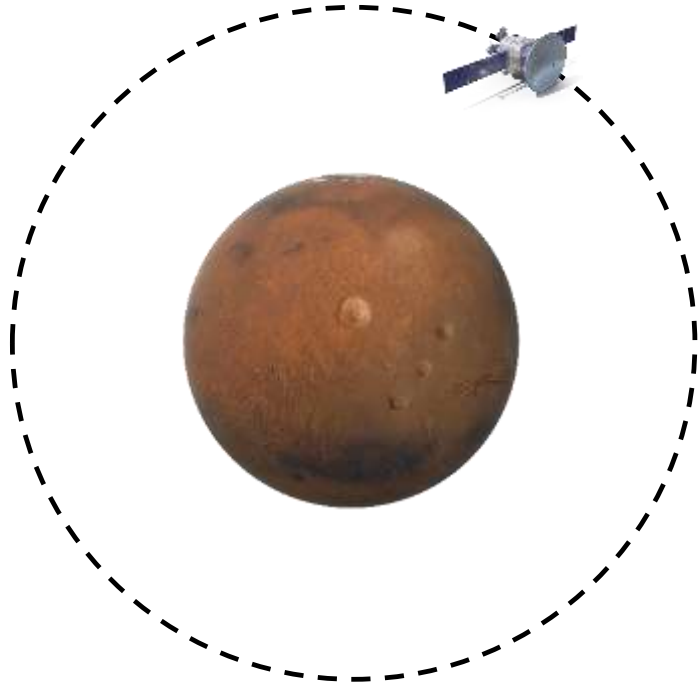
Options/Considerations for Relay Capabilities:

- If SEP is used, large solar panels can be used for high power relay needs
- Relay bandwidth could either be split between missions during the earth day or take multiple mission data rates at once
- May be able to rely on other infrastructure (MARCONI)

Options for Final Relay Orbit:

- Stay in final drop off orbit
- Kick out to high orbit for high coverage
- Remain in science orbit for hosted instruments if there are any

Space Tug Mission Drop Off – High Level Options



- “Bus Stop”
 - Tug performs orbit changes to drop each satellite off in rough desired orbit.
- “Carrier Deployment”
 - Tug will utilize an auxiliary carrier such as the ION Satellite Carrier from D-Orbit which can perform orbit changes and deploy CubeSats as well as potentially host instruments.
- “One Stop Shop”
 - Tug will enter one pre-determined orbit based on manifest and require all vehicles to provide the dv to reach desired science orbit from drop off.

Space Tug Strawman Mission

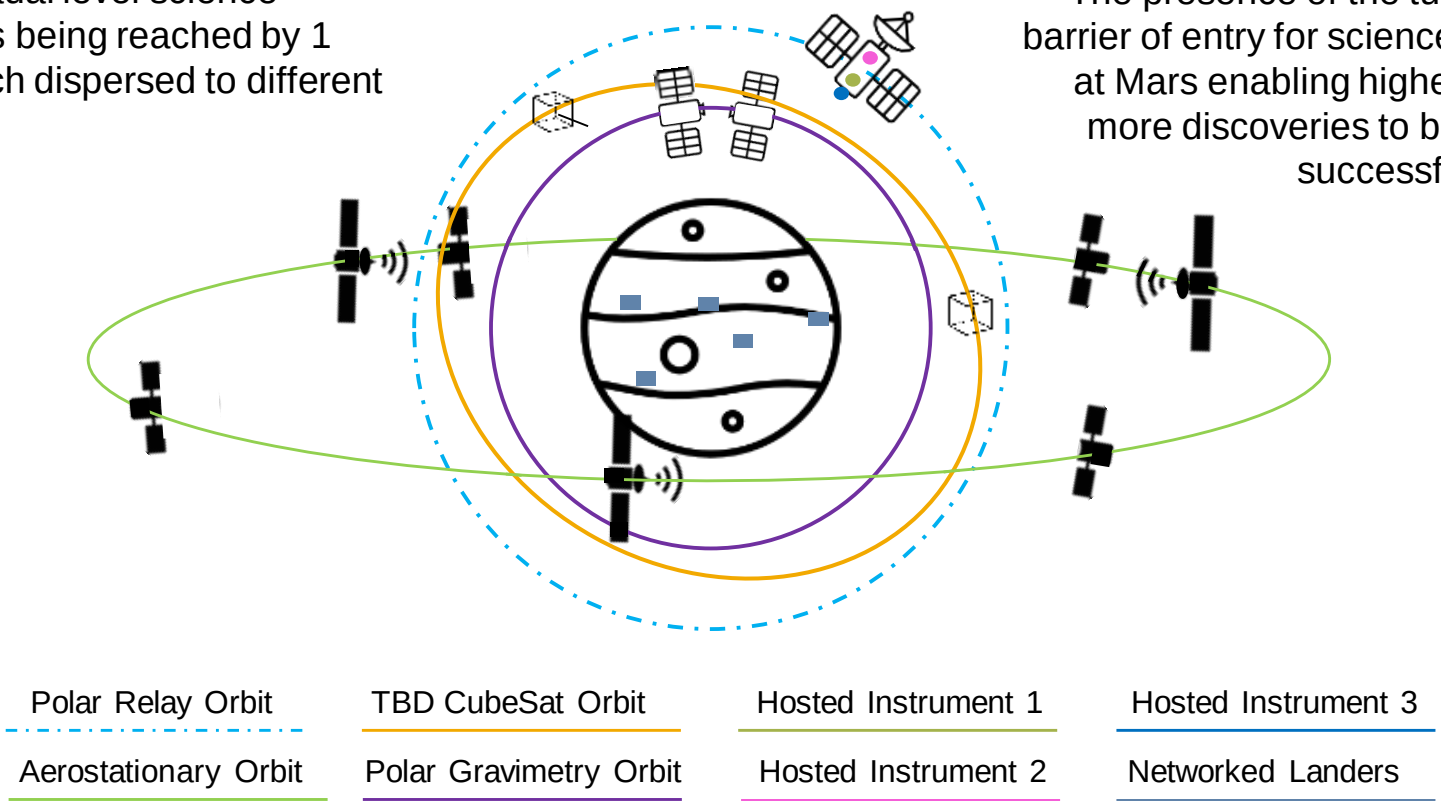
Potential Tug Manifest:

Mission	Science Orbit	Mass Class	# Vehicles
Atmospheric Optical CubeSat Instrument	Aerostationary	CubeSat	3
Space Weather Magnetometer	Aerostationary	CubeSat	3
Surface Penetrator	Surface; distributed	CubeSat	5
Meteomars atmosphere and climatology	Aerostationary	SmallSat	3
MAWI	TBD	Instrument	1
MARSSI	Near polar	Instrument	1
2 Element Polarimetry	TBD	Instrument	1
2 Element Polarimetry	TBD	CubeSat	1
Gravimetry	Near polar	SmallSat	2
Cosmic Ray Detector	Aerostationary	CubeSat	1

Potential Manifest Orbit Configuration

Each Decadal level science objective is being reached by 1 large launch dispersed to different orbits

The presence of the tug lowers the barrier of entry for science to be done at Mars enabling higher return and more discoveries to be made with successful execution



- | | | | |
|-----------------------------|-------------------------------|----------------------------|----------------------------|
| <u>Polar Relay Orbit</u> | <u>TBD CubeSat Orbit</u> | <u>Hosted Instrument 1</u> | <u>Hosted Instrument 3</u> |
| <u>Aerostationary Orbit</u> | <u>Polar Gravimetry Orbit</u> | <u>Hosted Instrument 2</u> | <u>Networked Landers</u> |

Conclusions

Summary of Contributions

1. Identified 2 high-desire and medium to high feasibility potential Mars SmallSat science mission concepts
2. Explored trade space for mission concept designs
3. Presented concepts to the International Mars Exploration Working Group (WG) Low Cost WG
4. Generated concept of operations for a potential space tug manifest inclusive of other international concept and one of mine
5. Contributed to understanding of accommodations required by a space tug in context of science missions
6. Identified areas for future analysis

Lessons Learned

- Mission Formulation forces design choices to be made and there can be incorrect choices.
- There is a lot of existing technology enabling small spacecraft science at Mars if given the right infrastructure.
- Certain missions, although focused, have widely reaching impacts on all science facets.
- Co-manifested missions can present opportunities for communication, theming, and “free” science through those communications.

Future Goals

- Study changes caused by additional radio occultation instrument
 - Additional mass and power needs contribute to prop systems and reaction wheels for orbiter
 - Additional mass and power needs may force instrument trades to accommodate 6kg soft limit for Lander
- Further tailor instrument box for lander with orientation and mechanical features in mind
 - Accounting for bounce and survival of extendable booms for certain sensors
- Study relay limits for tug given low or high load carried mission distributions
- Make concrete decisions on component make up of gravity orbiter
 - Attempt to remain within mass predictions and have a goal of minimal specialized components

Thank you!

Special Thank You to

Michael Fifield Marguerite Syvertson

Molly Goldstein Nina Andrukh

Steve Matousek Steve Zusack

Young Lee Robert Miller

Lyn Repath-Martos Mary-Lou Sutherland

Mars Exploration Program

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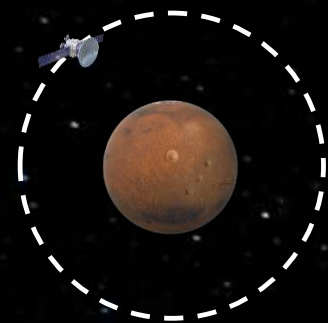
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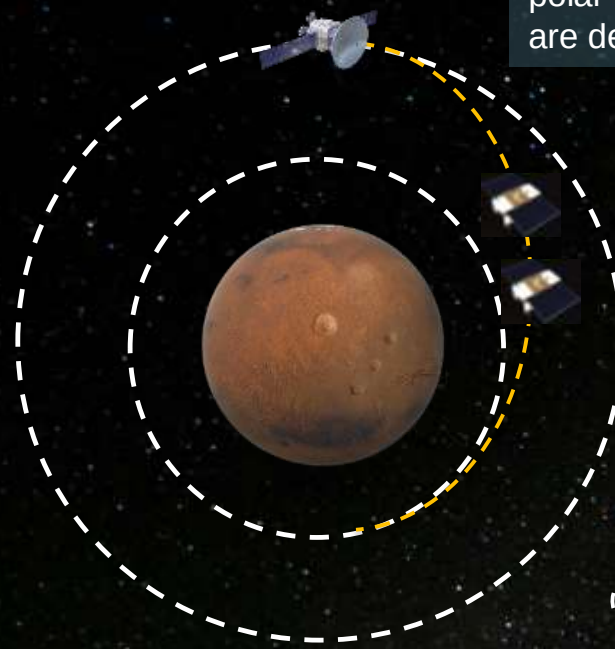
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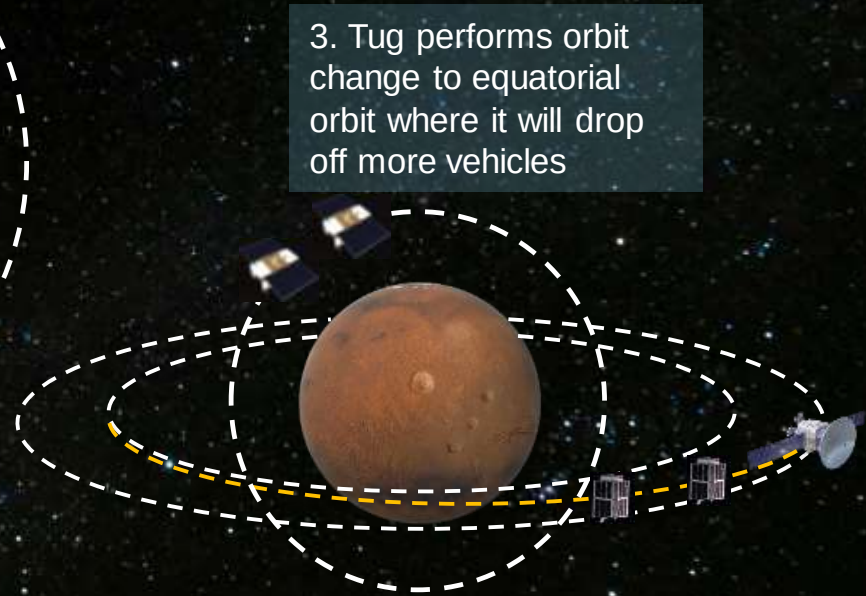
Bus Stop



1. Space Tug enters circular polar low Mars orbit



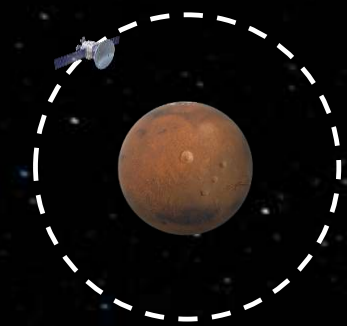
2. All vehicles requiring polar orbit deployment are deployed



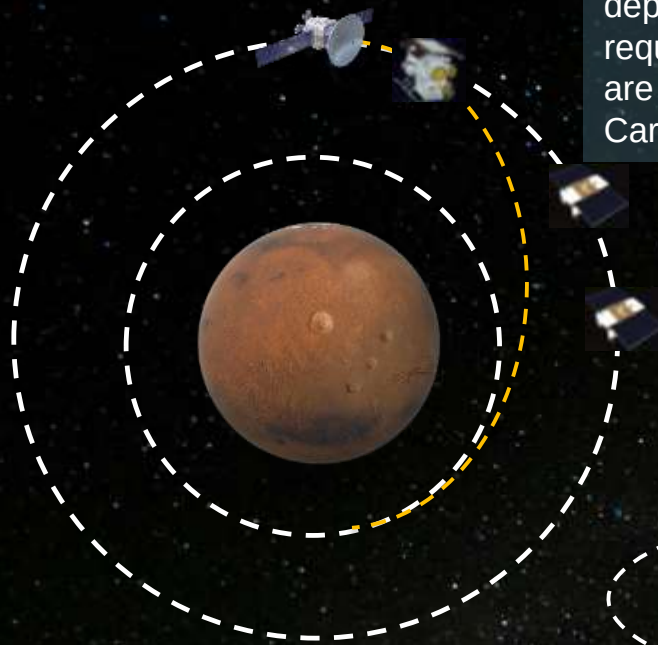
3. Tug performs orbit change to equatorial orbit where it will drop off more vehicles

4. Tug will transfer to relay orbit after dropping off remaining vehicles

D-Orbit Deployment

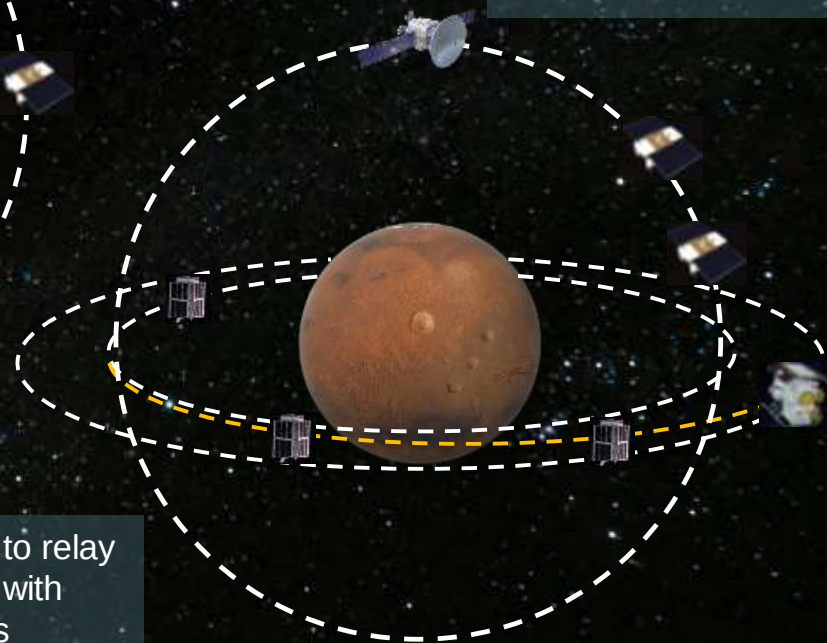


1. Space Tug enters circular polar low Mars orbit



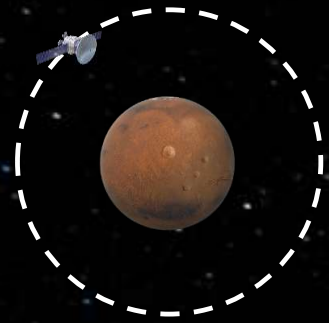
2. All vehicles are deployed, vehicles requiring furthest travel are deployed in ION Carrier

3. ION Carrier deploys farthest vehicles in CubeSats mass class

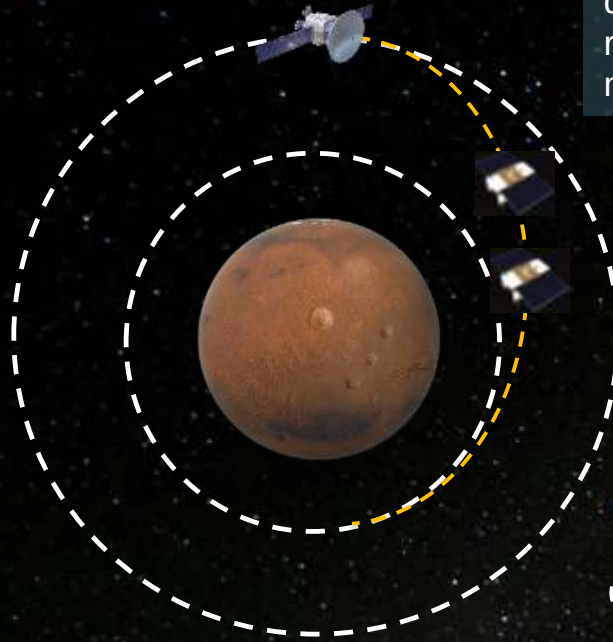


4. Tug will transfer to relay orbit once finished with polar orbit drop offs

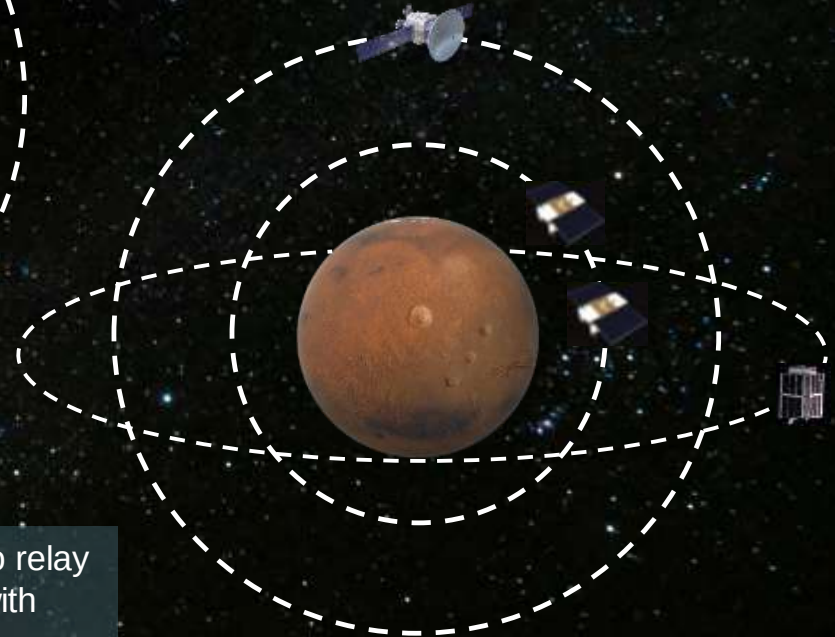
One Stop Shop



1. Space Tug enters circular polar low Mars orbit



2. All vehicles are deployed, vehicles requiring furthest travel must travel on their own



3. Tug will transfer to relay orbit once finished with drop offs